R15

Code No: 123BR

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, November/December - 2017
BASIC ELECTRICAL ENGINEERING
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) State the Superposition theorem. [2]
b) A conductor is 50 m long. It has a cross – sectional area of 2 mm^2 while it offers a resistance of 10 Ω. Find the conductivity of the material. [3]
c) How does inductance and capacitance will behave when an alternating sinusoidal voltage is applied to them. [2]
d) A coil has an inductance of 50 mH and negligible resistance. Find its reactance at 100 Hz. [3]
e) What are the conditions to be fulfilled to work as step up transformer? [2]
f) Distinguish between Ideal and a practical transformer. [3]
g) Explain the principle of operation of DC Generator. [2]
h) What is the difference between a separately excited and a self excited generator? [3]
i) List the advantages of Moving coil instruments. [2]
j) List the necessary purpose of providing the controlling torque in indicating Instrument. [3]

PART-B

(50 Marks)

2.a) State and explain Kirchhoff's current law and Kirchhoff's Voltage law.
b) Find the equivalent capacitance of the system across terminals a-b shown in the figure 1. [5+5]

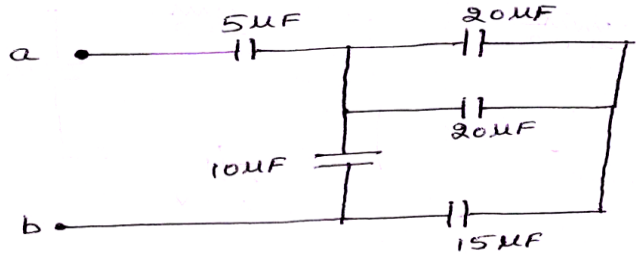


Figure: 1

OR

3.a) Explain the process followed for Star (Y) – Delta (Δ) transformation.
b) Find the Thevenin's equivalent for the network shown in figure 2 across R_L. [5+5]

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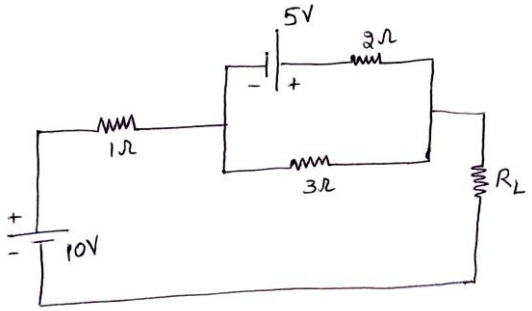


Figure: 2

4.a) How do you generate alternating emf and explain the terms phase angle and Amplitude.

b) Find the average and rms of the waveform shown in the figure 3. [5+5]

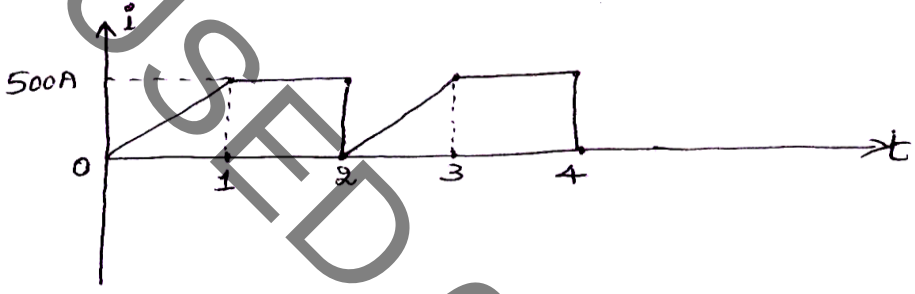


Figure: 3

OR

5.a) Explain the terms with respect to Alternating Quantity: i) Average value ii) Form factor.

b) A voltage of 400 V is applied to a series circuit containing a resistor, an inductor and a capacitor. The respective voltages across the components are 250 V, 200 V and 180 V and the current is 5 A. Determine the phase angle of the current. [5+5]

6.a) Explain the principle of operation of a transformer.

b) Define the term Regulation and derive the expression for regulation of a transformer. [5+5]

OR

7.a) Explain in detail about Core type and shell type transformers.

b) A 200 KVA single – phase transformer has 1000 turns in the primary and 600 turns on the secondary. The primary winding is supplied from a 440 V, 50 Hz source. Find the i) Secondary voltage at no load and ii) Primary and secondary currents at the full load. [5+5]

8.a) Distinguish between the short shunt and long shunt compound generators.

b) A 4 pole lap wound DC Generator has 680 conductors; a flux of 30 mWb per pole driven at 550 rpm. Find the generated emf. [5+5]

OR

9.a) Explain in detail about lap and wave windings in DC Machines.

b) A three phase, 50 Hz, 6- pole induction motor runs at 950 rpm. Calculate:
i) the synchronous speed ii) the slip and iii) frequency of the rotor emf. [5+5]

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10.a) Explain the different methods of generating Damping torque and explain them with neat sketches.

b) List the different types of errors that occur in Moving iron instruments and how can we compensate them. [5+5]

OR

11.a) Explain in detail about Indicating, Integrating and recording electrical instruments with an example of each type.

b) With a neat sketch explain the working of Repulsion type moving iron instrument. [5+5]

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